

SFF discussions

Agenda June 12, SFF phase 2

University of Oslo

June 12, 2026

Outline

- 1 Agenda
- 2 Localization
- 3 Budget
- 4 Planning of meetings
- 5 RCN feedback
- 6 Part 1: A less in-depth interpretation
- 7 Part 2: Explicit things we can address
- 8 Overall interpretation

June 12 agenda

- Short summary from previous meeting (June 5)
- Localization
- Budget (Lasse)
- Planning of next meetings and writing
- Discussion of evaluation (if time)
- AOB

Starting discussion about localization

- Lead suggestion: Justin and Thomas
- Estimate on how many we could be
- Need to get started as soon as possible

Potential students, CS Physics and Chemistry, approx 14-16

Retning	EU-søkere	EU-1.pri	EU-tilbud	EU-ja	Norden-søkere	Norden-1.pri
AMRA	16	3	0	0	62	16
Astro	10	3	0	0	17	7
Bio	21	10	2	1	38	14
Kjemi	2	1	1	1	11	6
Geo	8	3	1	1	25	10
IFI	12	1	0	0	50	14
Mekanikk	5	0	0	0	13	4
Fysikk	9	2	3	2	63	19
Total	83	23	7	5	279	90

Quantum, Mena and other, possibly 15-20

Studieprogram	Opptaksramme	Søknader	Søkere	1.prioritet
Data Science	20	312	312	148
Elektronikk, informatikk og teknologi	30	114	88	48
Entreprenørskap og innovasjonsledelse	25	337	337	140
Fornybare energisystemer	30	119	119	54
Fysikk	30	180	120	54
Kjemi	25	204	105	74
Kvantevitenskap og kvanteteknologi	15	47	47	23
Matematikk	25	98	79	39
Materialvitenskap for energi- og nanoteknologi	15	39	39	19
Nukleærteknologi	15	47	47	20
Stokastisk modellering, statistikk og risikoanalyse	15	219	188	74

PIs and associated researchers

Name	Title	Aff.	Role	RT1	RT2	RT3	RT4	Ext. funding
<i>Centre directors</i>								
Morten Hjorth-Jensen	Professor	Phys., UiO	Dir. ^a /PI	X	X	X	X	NSF, DOE, RCN
Marianne E. Bathen	Assoc. Prof.	Phys., UiO	Dir. ^b /PI	X		X	X	YRT ^c , Equinor
<i>Principal investigators</i>								
Thomas Bondo Pedersen	Professor	Chem., UiO	PI	X	X		X	RP ^d
Simen Kvaal	Researcher	Chem., UiO	PI	X	X		X	ERC-StG, CAS ^e
Michal Repiský	Sr. Researcher	Chem., UiO	PI	X	X		X	MSCA-IF ^f
Lasse Vines	Professor	Phys., UiO	PI	X		X		Toppforsk (RCN)
Justin W. Wells	Professor	Phys., UiO	PI	X		X		RCN
<i>Associated investigators (AI)</i>								
Susanne Viefers	Professor	Phys., UiO	AI	X	X			
Francesco P. Massel	Professor	Phys., UiO/USN	AI	X	X	X	X	RCN, QuantERA
Ainara Nova	Assoc. Prof.	Chem., UiO	AI					
Michele Cascella	Professor	Chem., UiO	AI				X	
David Balcells	Sr. Researcher	Chem., UiO	AI				X	
Erik Tellgren	Sr. Researcher	Chem., UiO	AI	X	X		X	
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Total estimate

- Pls, associated researchers and guests: 20
- Master of Science students: approx 30
- PhDs and PDs: estimate 15-20 at any time
- Total people: 65-70

Budget

- Lasse gives a brief intro and overview

When and plans for writing

- When and how often (zoom only most likely)
- First tentative draft to Innovayt week of July 8-13

Overall impression of the Phase 1 evaluation

- The review says this was a **very strong** Phase 1 proposal.
- All four criteria received the mark **6 (Excellent)** on the 1–7 scale. Is this scale correct?
- On the Research Council scale, mark 6 means the proposal *“successfully addresses essentially all aspects of the criterion and has only minor shortcomings.”*
- The path to a competitive Phase 2 application is therefore essentially about **closing the gap from 6 to 7 (Outstanding)** by addressing the specific *minor shortcomings* hinted at in each section.

Structure of these notes

- ① A less in-depth interpretation of the feedback.
- ② A discussion of explicit things we can address in the full proposal.

Excellence – Quality of R&D activities

This section contains the most concrete criticism. Two distinct points are flagged.

1. Research questions need sharper articulation

Each of the four tracks should open with a *small set of falsifiable scientific questions*, rather than thematic descriptions.

2. Machine-learning methods need more detail

The ML-guided control component is the methodological area the panel found least convincing. The full application should include:

- concrete algorithms and model architectures,
- training and data strategies,
- validation protocols,
- the interface between ML and the physics/chemistry tracks.

A dedicated methods section for ML-guided quantum control is likely warranted.

Impact

Two implicit concerns stand out.

Scale

The panel notes that the center is

relatively small compared with major international centers

This is the only comparative remark in the whole report – and it is not a compliment. Phase 2 should either

- grow the scope and team to match international peers, or
- explicitly argue why the chosen size delivers *more impact per krone* than larger competitors.

Generic impact narrative

The current text stays at the level of *application areas* without committing to specific deliverables, pathways, or partners. Phase 2 requires explicit dissemination, exploitation, and open-science plans, plus communication and engagement with different target audiences.

Implementation

some uncertainty exists regarding continuity and transition of leadership

Succession uncertainty is one of the recurring reasons SFF panels downgrade otherwise excellent centers.

Phase 2 should include a written succession plan

- Who can step in as director if needed?
- How are deputy and co-director roles structured?
- How is leadership of each of the four tracks backed up?
- How does the center survive the departure of a key PI?

Additional Phase 2 requirements (not yet present)

- An explicit training environment beyond what individual groups offer.
- Appropriate management and governance structures.
- A description of the physical organisation supporting cross-track collaboration.
- If gender-imbalanced: concrete plans to qualify under-represented-gender researchers for senior

How to read the comments

Our goal is to prepare a much larger and more competitive Phase 2 SFF proposal.

- We should read the comments not as criticism, but as indications of where reviewers want **additional depth, precision, and evidence**.
- I think that for highly competitive SFF evaluations, proposals with uniformly excellent scores are distinguished by how convincingly they address *these* details.

1. The clearest weakness: explicit research questions

The proposal would benefit from clearer articulation of the specific research questions within each track.

The reviewers liked the four-track structure but wanted more than broad themes.

Instead of:

- Quantum control of many-body systems
- Machine learning for quantum dynamics

Formulate questions such as:

- What are the fundamental limits on controllability in interacting many-body systems?
- Can agent-based AI (with reinforcement learning) discover control protocols beyond Pontryagin-optimal solutions?
- Can quantum Fisher information provide universal bounds on control fidelity?
- Can generative models compress experimentally relevant quantum states?

The best SFF proposals resemble a collection of major open problems, not activities.

2. Machine learning needs greater methodological precision

more detailed descriptions of certain methods – particularly those related to machine-learning-guided control.

The panel accepted that ML is important, but were not yet convinced it is sufficiently specified.

Name explicit ML methodologies:

- Agent-based AI and Reinforcement learning
- Physics-informed neural networks
- Bayesian optimization
- Diffusion models
- Transformer architectures
- Generative quantum state models
- Graph neural networks

For each method, explain:

- why it is needed,
- what scientific question it addresses,
- why existing methods are insufficient.

2. ML as a research theme, not just a tool

A common weakness in AI-related proposals is that ML appears as a *tool* rather than a *research theme*.

The proposal should argue

New machine-learning methodologies will emerge **from quantum control challenges themselves**.

Examples of methodological novelty

- Reinforcement learning for quantum control under uncertainty.
- Scientific foundation models for quantum experiments.
- Physics-informed generative models.
- Hybrid quantum-classical learning algorithms.

3. Clarify scientific ambition versus feasibility

The reviewers repeatedly praise the ambition, but Phase 2 should make the relationship between risk levels much more explicit.

Flagship challenges

- Fault-tolerant quantum control.
- AI-designed quantum sensors.
- Autonomous quantum experiments.

Fallback pathways

If a flagship goal fails:

- What is still learned?
- What publishable results remain?

SFF panels routinely ask: *What happens if the most ambitious ideas fail?*

4. Leadership succession and long-term sustainability

some uncertainty exists regarding continuity and transition of leadership

The panel implicitly asked:

- How dependent is the center on a single person?
- What happens after 5–10 years?
- How are younger leaders being developed?

Explicit leadership pipeline

- Deputy director(s).
- Future work-package leaders.
- Early-career leadership positions.
- Documented succession plans.

5. Demonstrate stronger center-level synergy

The next proposal must repeatedly answer:

Why does this need to be a Center of Excellence, rather than a collection of excellent groups?

Cross-track flagship projects

- Quantum sensing + AI.
- Quantum materials + machine learning.
- Quantum chemistry + control theory.
- Experimental platforms + digital twins.

Each major breakthrough should **require** collaboration between several tracks.

6. Strengthen international positioning

relatively small compared with major international centers

This reveals a comparison class. For Phase 2, compare explicitly to:

- QuICS (Maryland)
- IQIM (Caltech)
- IQC Waterloo
- CQT Singapore
- Munich Quantum Valley
- Quantum Delta NL

Then explain

- Where Norway can *lead*.
- What niche is uniquely Norwegian.
- Why the center will become internationally indispensable.

7. Expand the impact narrative

The current impact section is strong but generic (secure communication, advanced materials, healthcare, clean energy). Phase 2 should make this much more concrete.

Scientific impact

- New theory of quantum control.
- New algorithms.
- Open-source software.

Technology impact

- Quantum sensors.
- Quantum materials discovery.
- Quantum computing architectures.

Human capital impact

- Number of PhDs.
- Industrial internships.
- International mobility.

8. What is missing entirely from the criticism?

Interestingly, there are **no** criticisms regarding:

- scientific quality,
- leadership competence,
- publication record,
- feasibility,
- interdisciplinarity.

This is very encouraging. The panel is convinced the team can do the work.

The next proposal should focus less on proving excellence and more on:

- 1 Sharper scientific questions.
- 2 More detailed methodology.
- 3 Clearer ML strategy.
- 4 Stronger center-level integration.
- 5 Leadership succession.
- 6 A stronger international vision.

Reading between the lines

We are convinced that this is an excellent team pursuing an important and potentially transformative topic. For the full SFF proposal we would like to see a more detailed scientific roadmap, especially regarding machine-learning-guided quantum control, clearer articulation of the key scientific questions, and stronger evidence that the center will function as more than the sum of its individual research groups.

- A very favourable position going into a second-round SFF application.
- The reviewers did not identify fundamental weaknesses.
- They asked for **greater precision, sharper focus**, and a **more developed center strategy**.

Summary: priorities for the Phase 2 proposal

- ➊ **Leadership continuity and succession plan** – the only explicit Implementation concern.
- ➋ **Detailed methodology for ML-guided quantum control** – dedicated methods section.
- ➌ **Sharpened research questions** per track – falsifiable, ambitious, specific.
- ➍ **Scale and international positioning** – either grow, or justify the chosen scale against named peer centers.
- ➎ **Center-level synergy** – cross-track flagship projects that no single group could deliver.
- ➏ **Risk management** – flagship goals with explicit fallback pathways.
- ➐ **Phase 2 requirements** – dissemination, exploitation, open science, training environment, gender plans, governance.